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THE TWENTY-FOUR QUESTIONS

By Severin, of Chaen.

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Read before the First General Assembly of the Silver Society on its last day, in the Year of the Return of Radiance, C.R. 824, the year of the League's founding.

Friends and colleagues,

The Assembly rises today. Before it met, the Society charged me to gather into one place the hard questions that have passed these many years in letters between the cities and the schools, to set them in order, and to read them to you before we part. I have spent half a year upon the gathering. I struck out what was repeated, and where several letters asked the same thing I made of them one question; there remain twenty-four. Before I read them I would say a few words on why they are to be read at all, and why in this manner.

Our generation has done its learning for the most part upon books recovered. The War of the Saints lasted a hundred years; the libraries of many cities were burned and many copies scattered. In the twelve years since the war, you and I have gone from city to city seeking them, have pieced them together volume by volume, and have collated the fragments after the order that Babel left us. That work is not finished, and it must be finished. Yet I have often thought: our forebears kept the books that Babel left behind, but the old books we recover must one day all have been read through, and the blank pages where those before us never set pen must be filled by those who come after. What I shall read to you today are the places where we have looked in the old books and found no answer.

A learning is alive so long as it has questions to ask. When its questions are all answered, or when no one asks any more, nothing is left of it but copying. The use of a good question does not lie in its answer alone. To answer it men must devise new methods, and the new methods may be taken to answer other questions; the longer a question stands unanswered, the more methods, as a rule, grow up around it. There are many methods in the old books of Babel which we can use to this day without knowing for what question they were first devised.

Questions come by two roads. Some arise from what we see in the world: why the tides rise and fall, how a mana core is formed in the sea of cloud, why the plagues come most in the tide years. Others arise from the learning itself: whoever counts the primes is led, as he counts, to ask how they are spread further on; whoever writes down the laws is led, as he writes, to ask whether this manner of writing can write down every law. The first keep learning close to the things of the world; the second make learning grow new methods of its own. Both must be asked. Among the twenty-four there are both, and often a question of the first kind, pursued far enough, becomes one of the second.

As to what questions deserve a place in the list, I have held to five rules.

The first is that a question must be asked clearly; so clearly that, when one day a man brings an answer, we shall all be able to judge whether he has answered it. Each question is therefore first set down as a plain question. Some of them are still roughly put, because

I could put them no better. If those who come after can put them more finely, that is good; to put a question finely is already a step toward its answer.

The second is that a question must be hard, so hard that it cannot be answered without some new method; but not so hard that there is no way in. A question of which no one can even see where to begin serves only to dismay. To most of the questions, therefore, I have added a few words on how much we now know, and where one might begin. One or two have only their sentence, because beyond asking I have nothing to say.

The third is that "it can be done" is an answer, and "it cannot be done" is an answer also. To find out which roads are impassable is itself part of the work of learning. Among the books of Babel there is an old proof that there is no greatest prime. Once that proof was made, no one in the world went looking for the greatest prime again; this one "there is none" has saved those after us how much labour, and opened how many other questions! It may be that the answer to some of today's questions is "it cannot be done". If so, I pray you not to think the labour wasted: to prove that a thing cannot be done is, in the eyes of the Society, as much an answer as to prove that it can.

The fourth is that an answer must be written out, so that a scholar of another city can check it. A proof must be written so that it can be followed step by step; a proof that only its author can follow is not yet an answer. Where we ask about the things of the world, the answer must state the conditions and the errors of its trials, and men of another city, doing as it says, must obtain results that agree; a method that has worked every time it was tried, but of which no one can say under what conditions it works or how well, is still only a clue. A reading of a script must agree with the other fragments. In law, in number, and in casting, this rule is the same. Some say that a strict proof is always a long one, and that insight reaches the end sooner. I do not think so. He who writes strictly must make clear to himself by what right each step stands; when that is clear, the needless steps fall away of themselves, and the strict proof is often the simple one. Insight is a good thing, but it becomes learning only when it is written in words that others can check.

The fifth is that it is better to divide one question into several than to knead several matters into one. If a question hides several things within it, then when one of them has been answered others will suppose that all have been. Matters that belong together may stand in one question, but the parts that must be judged separately are to be marked.

When a question cannot be answered, two roads remain. One is to ask it more narrowly, and first answer some special case; it is in this sense that I ask the Third Question first of planes of three dimensions. The other is to ask it more widely: a question that cannot be answered by itself will sometimes show its way in when it is looked at among a whole kind of questions. If both roads fail, it is most often because some method is lacking, or some manner of writing. Our way of writing numbers has come down to us through thousands of years, and many matters of number become simple when they are written otherwise; it is the writing of the laws that the Sixth Question asks after.

I have besides a conviction, which I lay before you to weigh: that every question clearly asked has an answer, whether "it can be done", or "it cannot be done", or "it can be done under these conditions". That we cannot answer a question today does not mean that it can never be answered. This conviction is my own; the Society does not depend upon it in listing these questions. Only, he who says "it cannot be done" must say clearly what it

is that cannot be done, and under what conditions; to be unable to answer for a time is no proof that the thing cannot be done.

The order of the twenty-four runs, roughly, from the most abstract to what touches us most nearly. The first to the fifth ask about number, shape and proof; the sixth to the eleventh, about law and casting; the twelfth to the fourteenth, about the elements, the tides and the mana pivots; the fifteenth to the twentieth, about the cities, the crystals, the cores, translation and seals; the twenty-first to the twenty-third, about plague and the mana pool; the twenty-fourth, about the books of Babel. I do not mean by this order that the earlier are of more account. One learning has often to borrow the methods of another; I read them together that each of you may look at the questions of the others.

THE FIRST QUESTION. — OF THE DENSITY OF THE PRIMES

Question. As the primes grow sparser, is there an exact law of their density?

Between one and a hundred there are twenty-five primes; between a hundred and two hundred only twenty-one; and by ten thousand there remain on the average some twelve in every hundred numbers. The books of Babel prove that there is no greatest prime, but they do not say how the primes grow sparse. The copyists of Chaen have counted the primes below a hundred thousand for me, noting the count at every ten thousand; the counts are given in Table I.

Table I. The primes below one hundred thousand, as counted by the copyists of Chaen

Counted to	*Number of primes*	*Added in this ten thousand*
10,000	1,229	1,229
20,000	2,262	1,033
30,000	3,245	983
40,000	4,203	958
50,000	5,133	930
60,000	6,057	924
70,000	6,935	878
80,000	7,837	902
90,000	8,713	876
100,000	9,592	879

The primes added in each ten thousand grow fewer, but they grow fewer very slowly, and not steadily: the eighth ten thousand has some twenty more than the seventh. Looking at these numbers I judge that the primes thin out, on the whole, evenly; so evenly that it seems there must be a law, while the small rises and falls seem to have none. The primes are like raindrops falling on the ground: watch a small patch, and the drops fall without any order; look at a wide field, and they fall evenly.

I ask: given a number, how many primes are there below it, and is there an exact law? If no exact law can be written, how near can an approximate law come, and by how much at most can it be wrong? This question may begin with what is approximate, but it should not stop there. What I seek is not a greater table, for a table can always be counted further one number at a time; it is a law by which, without counting one by one, the density may be told as a whole, together with the most by which it may be wrong. I guess that the exact law will not be a simple formula; it may be that one must count in another manner,

counting not the primes themselves but something else joined with them, and reasoning back from that.

The copyists told me one thing more: the primes sometimes crowd together, and sometimes stand far apart; pairs of primes that differ by two, such as eleven and thirteen, seventeen and nineteen, are still met with often as far as a hundred thousand. Whether such pairs go on without end I do not make a question of its own, but set it down here for those who count the primes to bear in mind.

THE SECOND QUESTION. — OF EVEN NUMBERS AND PRIMES

Question. Is every even number greater than two the sum of two primes?

This question I copied from the margin of an old manuscript in Chaen. It is a book of number; who copied it no one knows. Between his pages of copying he amused himself by breaking even numbers into parts, and at the corner of the leaf he left this line in a small hand. I have tried it for him, and it holds for every even number below ten thousand; many even numbers can be broken in more than one way, and the larger numbers, on the whole, tend to have more ways. But I do not know why it holds, nor can I see how it might be proved. A question that can be asked so plainly in a single line, and that we cannot answer, is the very kind the Society should list.

This question looks near to the first, but it is different. The first asks how many primes there are on the whole; this asks whether every single even number can be broken exactly. To know roughly how many primes there are on the whole is not to know that every even number can be broken: a single even number that cannot be broken would make the line false, and what holds on the whole says nothing of any one number. That the larger numbers tend to have more ways makes one believe it true; but "more ways" holds on the average, and the average does not reach to every number.

THE THIRD QUESTION. — OF THE SHAPES OF CLOSED PLANES

Question. What shapes may a closed plane have?

There are demiplanes in which a man who walks straight on comes back, after a while, to the place he started from; they have no edge, and there is no walking out of them. I ask first of the simplest kind: planes finite in extent, all of one piece, without edge, and like ordinary space in the neighbourhood of every place. That a man comes back where he started is only one sign of such a plane, and not a test of it. What shapes may such a plane have? Given two closed planes, is there a way to judge whether they are of the same shape, and so to list all the shapes there can be? By shape I mean here not size but manner of joining: if one plane can be changed slowly into another without tearing, without joining, and without bringing two places into one, they are of one shape. Planes of three dimensions touch us most nearly, and I ask first of these; of more dimensions we cannot yet even say how we ought to think, but I list them here as well.

At present we know only one method: to draw a cord round some circuit in the plane and see whether it can be drawn together to a point without being cut. A plane in which every cord can be drawn together, and a plane in which some cord cannot, are surely not of the same shape. That method tells some shapes apart, but not all: whether two planes in each of which every cord can be drawn together must therefore be of one shape, I do not know.

THE FOURTH QUESTION. — OF THE FOUNDATIONS OF NUMBER

Question. Can the rules of derivation prove of themselves that they do not contradict themselves?

From accepted truths we derive, by a set of rules, the many conclusions of the science of number. Suppose that these rules can be listed one by one in due form, that they suffice to express the arithmetic of whole numbers, and that they do not in fact contradict themselves: can it then be proved, by these rules alone, that they will never lead both to a statement and to its contrary? We believe that they will not contradict themselves; but belief is not proof.

What this question asks is not that more conclusions be derived, but that the rules by which we derive them be given surety. A set of rules that has once led to a contradiction can lead to anything whatever, and all that it leads to is then worth nothing. It may be that this question requires first that derivation itself be written as something that can be checked by rule: which step is a derivation and which is not must be checked as an account is checked.

THE FIFTH QUESTION. — OF JUDGING A FORMULA TO ITS END

Question. Is there a means by which, before any formula is cast, its outcome may be known?

A formula, once unfolded, comes to rest or goes on working without end; whether a self-adjusting ward will go on adjusting itself for ever, we can now only cast it and watch; and that it is slow to stop is no proof that it will never stop. If there is such a means, many dangerous formulae need not be tried upon men; if there is not, we ought to know why not, and where the bounds of the "not" lie.

THE SIXTH QUESTION. — OF A RECKONING FOR THE LAWS

Question. Can the lines of law be written in a strict reckoning?

A law-mage sees the lines of law by what he grasped in ascending the Tower and by the divisions handed down in the manuscripts; when two of them speak of a line, they must often gesture at length, and when they have done they may not mean the same line. I ask whether the laws can be written as number is written, so that different men, following the reckoning, shall reach the same result. If they can, much that today can only be passed by word of mouth may be written down and checked.

The thirty-six thousand characters write how a working is to be cast, not how it is to be reckoned: with them we can carve a formula, but we cannot set two formulae side by side and reckon with them. The science of number travels far because it has a manner of writing, and what one man has derived another can derive again by following it. If the laws had such a manner of writing, they need not be handed down by insight alone.

THE SEVENTH QUESTION. — OF THE LAYERS OF THE WORLD

Question. By what way does a formula draw its power?

That elemental workings draw their power through the elements no one doubts. Whence law-workings and string-spells draw theirs, opinions differ. Some hold that the world has two layers only, the elements and Tiamat, and that the lines of law are Tiamat itself; then both kinds draw from the lines of law. Yet casters say often that within a ward

against magic a law-working cannot be cast, while a string-spell often can. How many layers lie between a formula and Tiamat, and what are they? Where the Babel fragments speak of a web, they use more than one word; I think there may be a clue there.

If this question can be answered, many strange things of casting may find their account: why the same formula cannot be cast in some places, and why some wards hold back one kind of formula and not another.

THE EIGHTH QUESTION. — WHETHER MANA IS CONTINUOUS

Question. Is mana continuous?

THE NINTH QUESTION. — OF THE BOUNDARY AT THE TENTH RING

Question. Why does casting harm the upper limit of the mana pool from the tenth ring on?

At the ninth ring and below it does not; only the Silver Folk take no such harm from any working below the thirteenth ring. Why does the common boundary lie at the tenth ring precisely, and that of the Silver Folk at the thirteenth? Is there a measure that relates the harm to the ring? Some say that it is the laws punishing those who exceed their right; that is a figure, not a reason. This question asks about the body of the caster and also about the frame of the lines of law, and I think the answer must be sought from both ends together.

THE TENTH QUESTION. — OF THE THIRTY-SIX THOUSAND CHARACTERS

Question. Do the thirty-six thousand characters suffice to write every law that can be written? And are there among them characters which, in every lawful use, can be spelt by others, and so are superfluous?

These characters came down to us from Babel. Here are two questions, which I have joined, because whoever answers the one will most likely have to think at the same time of the other. The two must be judged separately. If the characters do not suffice, there are laws that cannot be written in these characters, and more characters, or another manner of writing, will be needed; if some are superfluous, the carving may be made simpler, and less liable to error, by striking them out.

THE ELEVENTH QUESTION. — OF CAUSE IN WORKINGS OF TIME

Question. Can a working of time alter what had already happened before it was cast?

If it can, cause is no longer a line that runs forward, the records it touches must all be checked again, and how far its effects reach must first be made out; if it cannot, what is it that bounds it? This question may not be tried at the hazard of men; it must first be argued from the workings and the laws, and then sought in such observations as harm no one.

THE TWELFTH QUESTION. — OF THE TURNING OF THE ELEMENTS

Question. When the four elements are turned one into another, what is it that does not change?

Fire into air, water into earth: casters do it every day. Is there an account in which, entering the elements before and after, together with the vessels that hold them and all that lies about, the sums always agree? Today we know only that every turning has its

loss; we do not know where the loss goes. If there is such an account, the turning of the elements may be reckoned by it, and need not rest on experience alone.

THE THIRTEENTH QUESTION. — OF THE TIDES OF MANA

Question. Have the risings and fallings of the tides a fixed law?

The tides come once in ten years, and a great tide once in a hundred; this our forebears wrote down. Of the other risings and fallings, great and small, there seems no order. Have they to do with the twelve mana moons? Can they be reckoned beforehand, as the sun and moons are reckoned? This question wants the records of many generations. I ask every city to keep a record of its tides, and to keep it, as far as may be, in one manner: at what hours of the day to read, how many times, and with what instrument, should all be fixed; let each record also note the place of the reading, the day on which the instrument was last set true, and any unusual supply of mana at the time. Only when the records of the cities can be compared one with another can anything be reckoned from them.

THE FOURTEENTH QUESTION. — OF STEADY OUTPUT IN THE TIDES

Question. Can a mana pivot, under the shock of the tides, hold its own output steady?

When the tide comes, the output of a pivot rises and falls with it. Today all depends on the craftsmen who stand by the pivot, watching and adjusting. When the craftsman adjusts well, the city feels nothing; when he is slow, or adjusts too far, the supply comes now strong and now weak. The strength of men has its end: as the cities are built greater and the engines more numerous, a day will come when they cannot all be watched. So long as the tide does not exceed what the pivot can bear, can the pivot be made to hold itself steady, without a man to stand by it?

THE FIFTEENTH QUESTION. — OF THE KEEPING ALOFT OF THE CITIES

Question. When its levitation arrays are damaged in part, can a city keep from falling without the help of hands?

A city is borne upon several groups of arrays. If one group is damaged, the others must bear more; whether they can bear it rests today on the experience of the craftsmen, and if the repair is slow the city will lean. I ask whether, when the remaining arrays still have lift enough and the supply and the main ward hold, a city can keep itself steady and aloft without the help of hands.

THE SIXTEENTH QUESTION. — OF THE ELEMENTAL CRYSTALS

Question. When an elemental crystal has spent its mana, can it be renewed?

A spent crystal is a dead stone. If it can be renewed, by what?

THE SEVENTEENTH QUESTION. — OF THE MANA CORES

Question. When a mana core is spent, can it be charged anew?

If it can, is there loss in the charging, and how many times may it be charged?

THE EIGHTEENTH QUESTION. — OF DEFLECTED TRANSLATION

Question. Can the landing of deflected translation be reckoned beforehand?

Deflected translation has saved the lives of many law-mages, yet no one has yet reckoned its landing exactly, and sometimes it falls far off, into a place more perilous than

the one it fled. I ask whether the exact landing can be reckoned from what is known before the casting alone; and if it cannot, how small at least the error can be made.

THE NINETEENTH QUESTION. — OF THE BOUNDS OF SEALS

Question. Can the seals now in use be freed at last from tending?

All the seals now in use must be tended; every few years they must be searched and mended. Men are careless, and when a seal is lost it is the affair of a whole city. If they cannot be freed, by what law do they wear, and how often ought they to be searched?

THE TWENTIETH QUESTION. — OF THE MAKING OF MANA CORES

Question. How are the mana cores formed in the sea of cloud?

They are gathered from the sea of cloud, and each one gathered is one fewer. How long does a core take to form? Do we gather them faster than they grow? The sixteenth and seventeenth questions and this one are, as I see them, three faces of one matter: whether what is spent can be renewed, whether what is emptied can be filled again, and whether what is not yet there can grow. Together they ask how long the strength of the League can last.

THE TWENTY-FIRST QUESTION. — OF THE SOURCES OF PLAGUE

Question. Whence come the plagues?

There are several fierce plagues which strike the mortals of the cities, the people of mixed blood, the herds and the crops, while the Silver Folk do not take them. The old people say that the plague years are mostly tide years. Why do they come with the tide? Why do the Silver Folk not take them? That itself is a clue. When the source is known, a defence can be founded upon it; until it is known, the defences the cities have already found by experience should be recorded too. I ask every city to record together its tide years, its cases of sickness, and the traffic of its routes.

THE TWENTY-SECOND QUESTION. — OF THE NATURE OF THE MANA POOL

Question. What is the mana pool?

Where in the body does it lie, how does it grow, and how is it harmed? We know only that it may be drained and may recover, and that when its upper limit is harmed the harm is sometimes long in mending; but at present we cannot even tell a passing exhaustion from a lasting harm. So long as we cannot, we do not know which caster need only rest, and which must cast less and be treated.

THE TWENTY-THIRD QUESTION. — OF HARM TO THE MANA POOL

Question. Can damage to the mana pool be repaired safely, swiftly and for all?

I ask for a repairing that all may use, and not for the lucky recovery of one or two. This question can hardly be taken in hand until the twenty-second shows some light; I list it by itself none the less, because those who are harmed cannot wait.

THE TWENTY-FOURTH QUESTION. — OF THE UNKNOWN SCRIPT OF BABEL

Question. Can the third script in the scrolls of the Babel Relic be read?

In the scrolls of the Babel Relic, beside the two scripts that we can read, there is a third, which no one can read. The copyists copy it stroke by stroke as they copy the rest, only

they do not know what they copy. What does it say? I list this question because it may lead us back into the old books; but this time we must read them by methods of our own.

The twenty-four questions have been read. I have only a few words more.

These questions belong to several learnings, but they do not stand each alone. The methods by which the primes are counted may perhaps be used to count the risings of the tides; the judging of formulae and the foundations of number are two faces of one matter; and how the mana cores are formed and whether they can be charged again will most likely be answered from the same place. As learning is divided ever more finely, and each city and each school keeps its own field, this cannot be helped; but a question must often be answered by a man of another field.

Some fear that as learning grows ever greater a day must come when no one man can know the whole of it, and that learning will then fall apart. I do not see it so. Each new method makes many old things simple, and what once wanted ten volumes to understand will afterwards want one. Learning will grow ever greater, but it need not grow ever more confused.

The Society asks you to carry this list home, to copy it for the men of your city and your school, and to copy it also for those of other cities and other schools. Whoever answers a question, or proves that a question cannot be answered, is asked to send it to the Society, which will record it, and read it at the next Assembly.

This list is not sealed. Of the twenty-four, some will one day be asked more finely, some will be shown to have been asked amiss, and there will always be new questions to add. I hope only that at the next Assembly, and the one after, someone will stand here and read not questions but answers; and if what he reads is “it cannot be done”, that too will be an answer.

I have one request more. When a question is answered, let the answer be copied twice, and kept in two cities. We have only just brought our books back from a war, and you all know how easily a manuscript is lost.